

Ellipses and Hyperbolas: Reading a Model's Key Features

Answer Key

Algebra 2

Quick Answers

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|----------------------------|----------------------------|
| 1. 5 | 6. 6, 64 |
| 2. 6, 10 | 7. 2, 5.39 |
| 3. 8, -2, 7, -1 | 8. $m = 45$, $n = 20$, 5 |
| 4. $\frac{3}{4}$ | 9. $\frac{7}{4}$ |
| 5. $a = 7000$, $c = 2800$ | 10. $\frac{5}{6}$ |
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Curriculum

Level: Algebra 2

HSG-GPE.A.3 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 18 tagged checks.

Common wrong answers

9. If they got 0.57: divided the x -denominator by the y -denominator, which reverses the slope

1. Whispering gallery: $\frac{x^2}{169} + \frac{y^2}{144} = 1$ (feet). How far from the centre is each focus?

Solution: The larger denominator sits under x^2 , so the major axis is horizontal with $a^2 = 169$ and $b^2 = 144$. For an ellipse the focal relation is $c^2 = a^2 - b^2$ (subtract, because the foci lie inside the curve): $c^2 = 169 - 144 = 25$, so $c = \sqrt{25}$. Each focus is that far from the centre, one on each side.

$c = 5 \text{ ft}$

2. Radio towers: $\frac{x^2}{36} - \frac{y^2}{64} = 1$ (kilometres). Give the vertices and the foci.

Solution: The x^2 term is positive, so the transverse axis is horizontal: $a^2 = 36$ and $b^2 = 64$. The vertices are $(\pm a, 0)$ with $a = \sqrt{36}$.

$(\pm 6, 0)$

For a hyperbola the focal relation adds: $c^2 = a^2 + b^2 = 36 + 64 = 100$, so $c = \sqrt{100}$ and the foci are $(\pm c, 0)$.

$(\pm 10, 0)$

Sanity check: $c > a$, as it must be — the foci always sit beyond the vertices on a hyperbola.

3. Sprinkler patch: $\frac{(x-3)^2}{25} + \frac{(y+2)^2}{9} = 1$ (metres). Give the endpoints of the major axis, then the foci.

Solution: The centre is $(h, k) = (3, -2)$, read by flipping the signs inside the parentheses. The larger denominator, 25, sits under the x term, so the major axis is horizontal with $a = \sqrt{25} = 5$ and $b = \sqrt{9} = 3$. Step 5 metres right and left of the centre: $3 + 5$ and $3 - 5$, each paired with $y = -2$.

$$(8, -2)$$

$$(-2, -2)$$

Focal distance: $c^2 = a^2 - b^2 = 25 - 9 = 16$, so $c = 4$. Step 4 metres right and left of the centre along the same axis: $3 + 4$ and $3 - 4$.

$$(7, -2)$$

$$(-1, -2)$$

4. Cooling tower: $\frac{x^2}{144} - \frac{y^2}{81} = 1$ (metres). Find the asymptote slopes and equations.

Solution: Here $a = \sqrt{144} = 12$ and $b = \sqrt{81} = 9$. For a hyperbola opening left and right, the asymptotes through the centre have slopes $\pm \frac{b}{a} = \pm \frac{9}{12}$, which reduces.

$$\frac{3}{4}$$

The centre is the origin, so the two asymptote equations are $y = \frac{3}{4}x$ and $y = -\frac{3}{4}x$; together

$$y = \pm \frac{3}{4}x$$

5. Satellite orbit: closest approach 4200 km, farthest 9800 km, with the planet at one focus. Solve for a and c .

Solution: The closest point is the vertex nearer the focus, a distance $a - c$ away; the farthest is the opposite vertex, $a + c$ away. That is the system

$$a - c = 4200$$

$$a + c = 9800$$

Add the two equations to eliminate c : $2a = 14000$, so $a = 7000$ km.

$$a = 7000$$

Substitute back into the second equation: $7000 + c = 9800$, so $c = 2800$ km.

$$c = 2800$$

Check: $7000 - 2800 = 4200$ ✓ and $7000 + 2800 = 9800$ ✓.

6. Navigation stations: foci 20 km apart, distance difference 12 km. Find a , then b^2 .

Solution: On a hyperbola the constant difference of distances to the foci equals $2a$, so $2a = 12$ and $a = \frac{12}{2}$.

$$a = 6$$

The stations are the foci, so $2c = 20$ and $c = 10$. The hyperbola relation $c^2 = a^2 + b^2$ rearranges to $b^2 = c^2 - a^2 = \left(\frac{20}{2}\right)^2 - \left(\frac{12}{2}\right)^2 = 100 - 36$.

$$b^2 = 64$$

7. Hyperbolic mirror: vertices $(\pm 5, 0)$ cm, asymptotes $y = \pm \frac{2}{5}x$. Find b and c .

Solution: The vertices give $a = 5$. The asymptote slope of a left-right hyperbola is $\frac{b}{a}$, so

$$\frac{b}{5} = \frac{2}{5}, \text{ which means } b = 5 \cdot \frac{2}{5}.$$

$$b = 2$$

Then $c^2 = a^2 + b^2 = 5^2 + 2^2 = 29$, so $c = \sqrt{29} = 5.385 \dots$ centimetres, which rounds to two decimal places.

$$c \approx 5.39$$

8. Bridge arch: upper half of an ellipse centred at the origin through $(3, 4)$ and $(6, 2)$ (metres). Find $m = a^2$, $n = b^2$, and the foci.

Solution: Substituting each point into $\frac{x^2}{m} + \frac{y^2}{n} = 1$ gives

$$\begin{aligned}\frac{9}{m} + \frac{16}{n} &= 1 \\ \frac{36}{m} + \frac{4}{n} &= 1\end{aligned}$$

These are linear in $\frac{1}{m}$ and $\frac{1}{n}$. Multiply the second equation by 4 and subtract the first: $\frac{144}{m} - \frac{9}{m} = 4 - 1$, so $\frac{135}{m} = 3$ and $m = 45$. $m = 45$

Substitute into the first equation: $\frac{9}{45} + \frac{16}{n} = 1$ gives $\frac{16}{n} = \frac{4}{5}$, so $n = 20$. $n = 20$

Since $m > n$, the major axis is horizontal and $c^2 = m - n = 45 - 20 = 25$, so $c = 5$ metres. $(\pm 5, 0)$

9. Tunnel, error analysis: $\frac{y^2}{49} - \frac{x^2}{16} = 1$ (metres). Jae reports slopes ± 0.57 . Find the mistake and the correct slopes.

Solution: This hyperbola has the y^2 term positive, so it opens up and down and its transverse axis is vertical: $a^2 = 49$ sits under y^2 and $b^2 = 16$ sits under x^2 . For an up-down hyperbola the asymptotes are $y = \pm \frac{a}{b}x$ — the number from under y^2 goes on top. Jae used

$\frac{4}{7} = 0.57$, which is $\frac{b}{a}$: the slope was inverted by copying the left-right formula. The correct slope is $\frac{\sqrt{49}}{\sqrt{16}} = \frac{7}{4}$. $y = \pm \frac{7}{4}x$

Reading check: the tunnel is taller than it is wide near the vertex, so its walls must rise faster than 1 metre up per 1 metre across — a slope bigger than 1, not smaller.

10. Skate ramp: $\frac{x^2}{36} - \frac{y^2}{25} = 1$ (metres). Asymptote slope, sketch, and explanation.

(a) $a = \sqrt{36} = 6$ and $b = \sqrt{25} = 5$, so the slopes are $\pm \frac{b}{a}$. $\frac{5}{6}$

(b) Model sketch: mark the vertices at $(\pm 6, 0)$; draw the central rectangle with corners $(\pm 6, \pm 5)$; draw the two diagonals of that rectangle all the way out — those are $y = \pm \frac{5}{6}x$; then draw one branch opening right from $(6, 0)$ and one opening left from $(-6, 0)$, each hugging the diagonals as it goes.

(c) Model explanation (judge by reading): solving the equation for y gives $y = \pm \frac{5}{6}\sqrt{x^2 - 36} = \pm \frac{5}{6}x\sqrt{1 - \frac{36}{x^2}}$. As $|x|$ grows, $\frac{36}{x^2}$ shrinks toward 0, so the square-root factor creeps up toward 1 and the branch creeps up toward $\pm \frac{5}{6}x$. It never arrives: the factor is always strictly less than 1 because $\frac{36}{x^2}$ is never 0 for a finite x , so the branch stays strictly inside its asymptote. Equivalently, a point on the asymptote would force $x^2/36 - y^2/25 = 0$, not 1.